Road-Condition-Aware EV Range Prediction

Parth Dembla(23110234) , Shriniket Behera(23110306)
Instructor : Prof. Pallavi Bharadwaj

ES665:Advanced Transportation Electrification Technology

Abstract

•Accurate range estimation is essential for reliable electric vehicle (EV) operation, **trip planning, placement of chargers and improving driver confidence** during long-distance commute.
•Conventional SOC-based methods for range estimation ignore dynamic effects like **regenerative braking effects and road conditions, leading to inaccurate and inconsistent range predictions**.
•We propose a **machine learning approach using an LSTM network** that leverages time-series vehicle data such as **speed, acceleration, SOC, road type, and slope**.
•The model predicts **regenerative energy recovery and improves dynamic EV range estimation** across different driving conditions such as urban traffic,highways,slope terrains.

Motivation and Introduction

•Electric vehicle (EV) range estimation is critical for **driver confidence and trip planning**.
•Current EV range prediction systems rely primarily on **State of Charge (SOC)** and **average energy consumption models.** These models assume fixed driving cycles which don't reflect real patterns.

Problem Statement:

•Existing range estimation methods **ignore regenerative braking dynamics**
•Road c**onditions strongly influence regenerative energy recovery**[2]:
- Urban roads → frequent braking → higher energy recovery
- Highways → steady speed → minimal regeneration
- Downhill roads → sustained braking → large energy recovery

•**Data-driven modeling of regenerative braking is needed** to capture the impact of driving behavior and road conditions on EV energy recovery.

Objective:

- Develop a **machine learning model** that predicts **regenerative energy recovery and effective EV range** by incorporating:
  1. Vehicle dynamics
  2. Road conditions
  3. Temporal driving behavior.

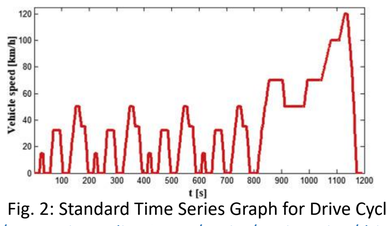


Fig. 2: Standard Time Series Graph for Drive Cycle
https://www.sciencedirect.com/topics/engineering/driving-cycle

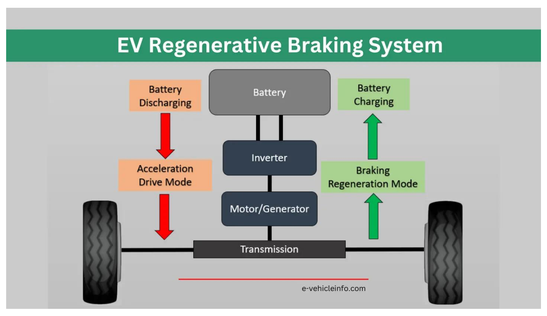


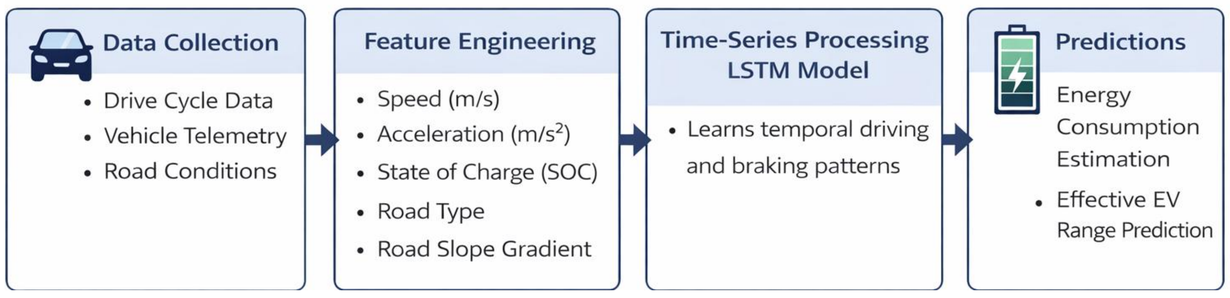
Fig. 1:Regenerative Braking System
https://e-vehicleinfo.com/regenerative-braking-system-in-electric-vehicles/

Literature Review

- Conventional SOC-Based Range Estimation[1]: $\text{Range} = \frac{SOC \times Battery\ Capacity\{KWh\}}{Energy\ Consumption\ Rate\{\frac{KWh}{Km}\}}$
- ML models have been used to estimate **EV energy consumption**.[3]
- Driving pattern identification improves **range prediction accuracy**.[4]
- However, **road-condition-aware regenerative braking modeling remains underexplored**.

- Driving behavior and road conditions significantly affect **energy recovery potential**, motivating a **data-driven time-series approach**.

Methodology



Why LSTM?

- Driving behavior is **sequential**.
- Key advantages:
  - Captures **temporal dependencies in driving patterns**
  - Maintains **memory of past braking events**
  - Learns **nonlinear relationship between vehicle dynamics and regeneration efficiency**

Data Collection

Dataset Used:

We use the **Vehicle Energy Dataset (VED / eVED)**, a large-scale dataset developed for studying **energy consumption and driving behavior in electric vehicles under real-world conditions**.[5]

• The dataset was created by collecting **vehicle telemetry and GPS data from electric vehicles during multiple real driving trips**.
• Recorded vehicle parameters include **speed, road slope, latitude, longitude, and instantaneous energy consumption** measured at regular time intervals.
• Additional road-context attributes such as **road elevation profiles, speed limits, and nearby intersections** were obtained using **map-based APIs and GIS tools**.
• This **time-series dataset captures realistic driving patterns**, enabling data-driven modeling of **EV energy consumption under different road conditions and driving scenarios**.

Data Cleaning & Preprocessing:

- Computed **road gradient from elevation data** to capture the effect of uphill and downhill driving conditions.
- Performed **feature correlation analysis** and retained the most relevant variables: **Speed (m/s), Acceleration (m/s²), Gradient, Elevation, and Speed Limit (km/h)**.

- Removed **missing or corrupted sensor readings** to ensure reliable training data
- Filtered **outliers in speed, acceleration, and energy consumption** caused by sensor noise or logging errors.

- **Speed shows the strongest correlation with energy consumption (0.37)**, indicating that vehicle speed significantly influences EV energy usage.

- **Acceleration also contributes to energy consumption (0.27)**, reflecting the impact of driving behavior on energy demand.

- **Speed limit is moderately correlated with speed (0.47)**, suggesting that road regulations influence driving patterns and energy usage.

- **Road gradient and elevation show weaker correlations**, but they still capture the effect of terrain on vehicle energy consumption.

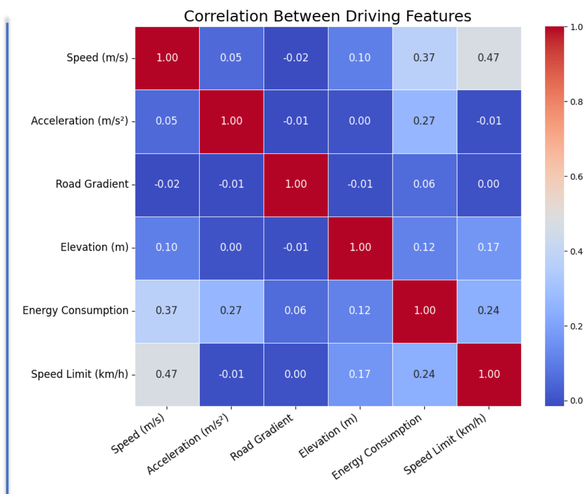


Fig. 3: Correlation matrix of selected driving features used.

LSTM(Long Short-Term Memory) Based Energy Consumption Model

- Driving behavior is **sequential in nature**, where past speed, acceleration, and road conditions influence current energy consumption.
- To capture these **temporal dependencies**, we use a **Long Short-Term Memory (LSTM) neural network** for time-series modeling.

Model Architecture

- **Input Layer:** Time-series sequence of selected driving features.
- **LSTM Layers:**
  - 2 stacked LSTM layers
  - **64 hidden units**
  - **Dropout = 0.2** to reduce overfitting

- **Fully Connected Layers**:
  - Linear layer: **64 → 32 neurons**
  - ReLU activation
  - Output layer: **32 → 1 neuron**

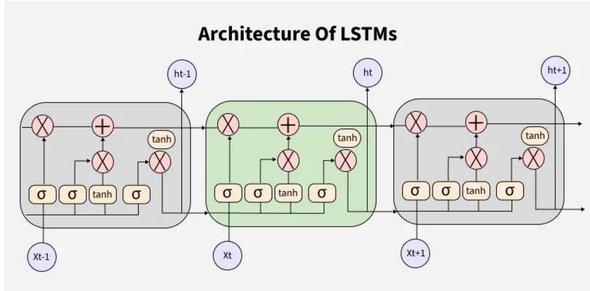


Fig.4: LSTM Architecture
https://media.geeksforgeeks.org/wp-content/uploads/20250528172141936296/architecture_of_lstms.webp

- **Output :**Predicted **energy consumption value** for the given driving sequence.
- The model uses the **final hidden state of the LSTM sequence** to summarize past driving behavior and predict energy consumption.

Initial Results

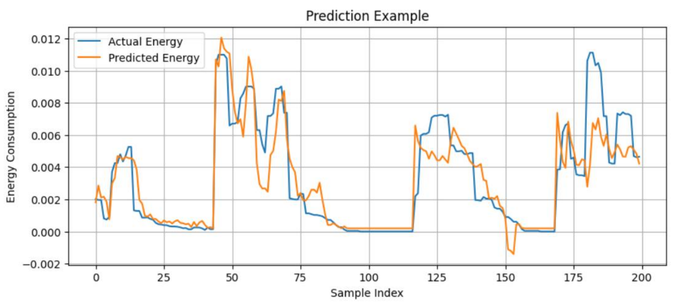


Fig.5: Actual & Predicted Energy Consumption

Sample Index: Sequential prediction step in the test dataset. Each index represents one LSTM prediction based on the previous **20 driving measurements**.

Test Results

- **Moderate Predictive Performance:** The model achieves an **R^2 score of 0.664**, meaning it explains about **66% of the variation in vehicle energy consumption across different driving conditions**. While the results demonstrate the model's ability to capture important temporal patterns, **there is still room for improvement.**

Plan Ahead

- **Feature Engineering:** Improve input features through systematic selection and derived driving metrics.[till **mid March**]
- **Extended Training:** Train for more epochs using improved configurations to further reduce prediction error. .[till **March end]**
- **Model Enhancement:** Evaluate advanced architectures such as **Stacked LSTM and BiLSTM**.
- **Hyperparameter Tuning:** Optimize learning rate, hidden units, and batch size[till **April first week]**

Future Scope: If the model is fairly accurate , then extend the model to Indian driving conditions and integrate it with map-based features (traffic, elevation, and road type) to predict route-level energy consumption and recommend the **most energy-efficient path based on the vehicle's remaining battery capacity.**

References

1. [1]H. Wei, X. Wang, and Y. Li, "Online estimation of driving range for battery electric vehicles based on energy consumption modeling," *Energy Reports*, vol. 8, pp. 1206–1217, 2022.
2. [2] J. Kropiwnicki and T. Gawłas, *"Energy efficiency of a car driving with regenerative braking," Combustion Engines*, vol. 202, no. 3, pp. 20–26, 2025, doi:10.19206/CE-207152
3. [3] Artificial intelligence and machine learning in optimizing EV powertrains," *Electronics*, vol. 12, no. 2, 2023, doi: 10.3390/electronics12020445.
4. [4] H. Yu, F. Tseng, and R. McGee, "Driving pattern identification for EV range estimation," *IEEE Trans. Vehicular Technology*, vol. 60, no. 7, pp. 3003–3015, Sep. 2011
5. [5] H. C. Oh, D. J. Leblanc and H. Peng, "**Vehicle Energy Dataset (VED), a Large-Scale Dataset for Vehicle Energy Consumption Research**," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 4, pp. 3302–3312, Apr. 2022, doi: 10.1109/TITS.2020.3035596.